

1. Database Objective

Database Name: FCCSS — Farmer Cooperative Crop Supply System

The FCCSS database is designed to digitize and manage the complete supply chain of agricultural produce — from farmer crop registration, through cooperative society processing and quality control, to final delivery to retailers and wholesalers. The system tracks government subsidies disbursed to farmers and manages logistics partners who transport crops from farms to cooperative warehouses.

Key business activities covered:

- Farmer registration and crop listing
- Cooperative society management : quality checks, procurement, pricing, and distribution
- Logistics management : vehicle allocation and farm-to-warehouse delivery tracking
- Retail and wholesale order fulfilment
- Government subsidy tracking : farmer-side input subsidies only
- Financial reporting : profit/loss per crop lot, payments to farmers, commissions

2. Application Users

- Farmers : Register crops, view procurement offers, track payments and subsidy credits.
- Cooperative Society Administrators : Core users. Manage procurement, quality checks, inventory, pricing, and distribution to buyers.
- Logistics Partners : Transport crops from farms to cooperative warehouses only.
- Retailers : Purchase processed/graded crops from the cooperative in smaller quantities.
- Wholesalers : Purchase large volumes of crops directly from cooperative stock.
- Government / Subsidy Authority : Records and tracks farmer input subsidy schemes and disbursements.
- System Admin : Manages user accounts, system configuration, and generates high-level reports.

3. Use Cases & Data Requirements

3.1 Farmers

Use Case	Description	Data Captured
Register Farmer	Farmer registers on the system with personal and farm details	Farmer ID, Name, DOB, Phone, Aadhaar/ID No., Bank Account, Village, District, State, PIN
Register Crop Lot	Farmer lists a harvest they wish to sell via the cooperative	Lot ID, Crop Name, Variety, Season (Kharif/Rabi/Zaid), Harvest Date, Estimated Qty (kg)
View Procurement Offers	Farmer views price offers made by cooperative for a crop lot	Offer ID, Offered Price per kg, Quality Grade, Cooperative ID, Offer Date
Track Payment	Farmer views payment status for sold lots	Payment ID, Lot ID, Amount, Payment Date, Mode, Status (Pending/Paid)
View Subsidy	Farmer views government input subsidies credited to their account	Subsidy Scheme ID, Scheme Name, Amount, Crop Type, Disbursement Date, Status

3.2 Cooperative Society

The Cooperative Society is the central entity. It procures crops from farmers, performs quality checks and grading, maintains warehouse inventory, and sells to retailers and wholesalers.

Use Case	Description	Data Captured
Register Cooperative	A cooperative society registers itself on the platform	Cooperative ID, Name, Registration No., Address, District, State, Contact Person, Bank Details
Farmer Membership	Cooperative enrolls farmers as members	Member ID, Farmer ID, Cooperative ID, Membership Date, Membership Fee, Status (Active/Inactive)
Procure Crop Lot	Cooperative makes a procurement offer and finalizes purchase from a farmer	Procurement ID, Lot ID, Farmer ID, Qty Procured (kg), Price per kg, Total Amount, Date
Quality Check & Grading	Cooperative inspects and grades each procured lot	QC ID, Lot ID, Inspector Name, Moisture Content (%), Grade (A/B/C/Rejected), Remarks, QC Date
Warehouse Inventory	Cooperative tracks stock of procured and processed crops	Inventory ID, Warehouse ID, Crop Name, Variety, Grade, Qty Available (kg), Received Date, Shelf Date
Set Selling Price	Cooperative sets price for each grade of crop for selling to buyers	Price ID, Crop Name, Grade, Price per kg, Currency, Effective Date, Set by (Admin ID)
Record Sale to Buyer	Cooperative records outgoing sale to a retailer or wholesaler	Sale ID, Buyer ID, Buyer Type, Crop, Grade, Qty, Price per kg, Total, Date, Delivery Status
Record Financials	Cooperative logs profit/loss per procurement-sale cycle	Finance ID, Crop Lot, Procurement Cost, Processing Cost, Selling Revenue, Profit/Loss, Period
Process Farmer Payment	Cooperative disburses payment to farmer after deducting costs	Payment ID, Farmer ID, Lot ID, Gross Amount, Deductions, Net Payable, Payment Date, Mode

3.3 Logistics Partners (Farm to Warehouse Only)

Logistics partners transport crop lots from farms to cooperative warehouses . The cooperative handles all subsequent delivery to buyers directly.

Use Case	Description	Data Captured
Register Partner & Vehicle	Logistics partner registers along with their vehicle details	Partner ID, Name, Contact, Service Region, GST No., Bank Details; Vehicle ID, Reg. No., Type (Truck/Mini-truck/Tractor), Capacity (kg)
Accept Transport Job	Partner accepts farm-to-warehouse crop movement request from cooperative	Job ID, Partner ID, Vehicle ID, Source (Farm), Destination (Warehouse), Crop Lot ID, Scheduled Date
Update Delivery Status	Partner updates the transport job status	Job ID, Status (Assigned/InTransit/Delivered/Failed), Actual Delivery Date, Remarks
View Payments	Partner views transport payments due from cooperative	Payment ID, Job ID, Amount, Payment Date, Status

3.4 Retailers & Wholesalers

Use Case	Description	Data Captured
Register	Retailer or Wholesaler registers on the platform	Buyer ID, Type (Retailer/Wholesaler), Business Name, Contact, Address, GSTIN, Bank Details
View Inventory	Buyer views available crop stock at cooperative	Crop Name, Variety, Grade, Quantity Available, Price per kg
Place Order	Buyer places a purchase order	Order ID, Buyer ID, Crop Name, Grade, Quantity, Delivery Address, Preferred Delivery Date
Track Order	Buyer tracks status of their order	Order ID, Status, Expected/Actual Delivery Date
View Invoices	Buyer views purchase invoices and payment history	Invoice ID, Order ID, Amount, Tax, Total, Payment Status

3.5 Government / Subsidy Authority

The government entity manages farmer-side input subsidy schemes (seeds, fertilizers, equipment).

Use Case	Description	Data Captured
Define Subsidy Scheme	Government creates a new farmer input subsidy scheme	Scheme ID, Name, Description, Type (FarmerInput), Eligible Crops, Amount per Unit or Flat Amount, Start Date, End Date
Disburse to Farmer	Government approves and disburses input subsidy to eligible farmers	Disbursement ID, Scheme ID, Farmer ID, Crop Lot ID, Amount Disbursed, Date, Reference No.
Subsidy Audit Report	Government views subsidy utilisation reports	Scheme ID, Total Disbursed, No. of Beneficiaries, Crop-wise breakup, Date Range

4. Key Entities & Attributes

Entity	Core Attributes
Farmer	FarmerID, Name, DateOfBirth, Phone, AadhaarNo, BankAccountNo, IFSC, Village, District, State, PINCode, RegistrationDate, Status
CropLot	LotID, FarmerID, CropName, CropVariety, Season, HarvestDate, QtyKg, Status (Listed/Procured/Rejected)
CooperativeSociety	CoopID, Name, RegistrationNo, Address, District, State, ContactPerson, Phone, BankDetails
FarmerMembership	MemberID, FarmerID, CoopID, EnrollmentDate, MembershipFee, Status
Procurement	ProcurementID, CoopID, LotID, FarmerID, QtyProcuredKg, PricePerKg, TotalAmount, ProcurementDate, PaymentStatus
QualityCheck	QCID, ProcurementID, InspectorName, MoisturePercent, GradeAssigned, Remarks, QCDate
Warehouse	WarehouseID, CoopID, Location, CapacityKg, Address
Inventory	InventoryID, WarehouseID, CropName, Variety, Grade, QtyAvailableKg, ReceivedDate, ShelfDate

Entity	Core Attributes
CropPrice	PriceID, CoopID, CropName, Grade, PricePerKg, EffectiveDate, SetByAdminID
Buyer (Retailer/Wholesaler)	BuyerID, Type, BusinessName, ContactPerson, Phone, Email, Address, GSTIN, BankDetails, RegistrationDate
Order	OrderID, BuyerID, CoopID, CropName, Grade, QtyOrdered, DeliveryAddress, OrderDate, Status (Pending/Confirmed/Dispatched/Delivered/Cancelled)
OrderItem	OrderItemID, OrderID, InventoryID, QtyKg, PricePerKg, TotalAmount
LogisticsPartner (incl. Vehicle)	PartnerID, Name, ContactPerson, Phone, ServiceRegion, GSTIN, BankDetails, VehicleID, VehicleRegNo, VehicleType, CapacityKg
TransportJob	JobID, PartnerID, VehicleID, JobType (Farm-to-Warehouse), SourceAddress, DestinationAddress, CropLotID, ScheduledDate, ActualDeliveryDate, Status, AmountPayable
SubsidyScheme	SchemeID, SchemeName, Description, Type (FarmerInput), EligibleCrops, AmountPerUnit, FlatAmount, StartDate, EndDate, GovernmentDeptName
FarmerSubsidyDisbursement	DisbursementID, SchemeID, FarmerID, LotID, AmountDisbursed, DisbursementDate, ReferenceNo, Status
Payment (to Farmer)	PaymentID, FarmerID, ProcurementID, GrossAmount, Deductions, NetPayable, PaymentDate, Mode, Status

5. Reporting Queries (All Roles)

#	Role	Query / Report	Purpose
1	Farmer	My crop lots and their current status	Track which lots are listed, procured, or rejected
2	Farmer	Payment history for all my sold lots	Check received and pending payments
3	Farmer	Subsidy amounts credited to me, scheme-wise	View government subsidy breakup
4	Farmer	Quality grades received on my submitted crop lots	Understand grading to improve next harvest
5	Farmer	Procurement price comparison across seasons for my crops	Decide best time to sell
6	Cooperative	Daily/Weekly/Monthly procurement summary (crop-wise, farmer-wise)	Monitor incoming stock
7	Cooperative	Current warehouse inventory — grade-wise and crop-wise	Stock management
8	Cooperative	Profit/Loss report per crop lot or per season	Financial health tracking
9	Cooperative	List of orders pending dispatch	Fulfilment operations
10	Cooperative	Undelivered orders with aging (days since order)	Escalation management
11	Cooperative	Total amount payable to farmers (crop-wise)	Payment planning

#	Role	Query / Report	Purpose
12	Cooperative	Sales summary — buyer-wise, crop-wise, region-wise	Sales analytics
13	Cooperative	Logistics cost breakdown per crop lot	Cost analysis
14	Cooperative	Quality rejection rate — crop-wise and farmer-wise	Quality KPI tracking
15	Cooperative	Seasonal price trend for each crop grade sold	Pricing strategy
16	Retailer/Wholesaler	My order history with delivery status	Track purchases
17	Retailer/Wholesaler	Current available crop inventory at cooperative (grade & price)	Browse before ordering
18	Retailer/Wholesaler	My invoices and payment outstanding	Financial reconciliation
19	Retailer/Wholesaler	Price history for a specific crop and grade	Price negotiation reference
20	Government	Total subsidy disbursed — scheme-wise, crop-wise, district-wise	Scheme utilisation audit
21	Government	List of farmers who received subsidy and amounts	Beneficiary tracking
22	Government	Farmers with no subsidy received (eligible but un-disbursed)	Coverage gap analysis
23	Government	Year-over-year subsidy comparison per crop type	Budget planning
24	System Admin	All users registered, by role and district	User management
25	System Admin	Cooperative-wise performance: procurement vs sales vs profit	Platform-level analytics
26	System Admin	Logistics partner performance: jobs completed, delays	Partner evaluation
27	System Admin	End-to-end crop journey: Farm → QC → Warehouse → Buyer	Supply chain audit
28	System Admin	Top 10 crops by volume traded on the platform	Market intelligence

6. Entity Interactions Summary

Entity A	Entity B	Relationship / Interaction
Farmer	CooperativeSociety	Farmer sells crops to cooperative; cooperative holds membership of farmers
Farmer	CropLot	Farmer registers one or more crop lots per season
CropLot	Procurement	Procurement is raised for each crop lot accepted by cooperative

Entity A	Entity B	Relationship / Interaction
Procurement	QualityCheck	Every procured lot undergoes mandatory quality check and grading
CooperativeSociety	Warehouse	Cooperative stores graded crop inventory in its warehouses
Inventory	Order	Buyer orders are fulfilled from warehouse inventory
Order	Buyer	Retailer or Wholesaler places one or more orders
CropLot	TransportJob	Crop transported from farm to cooperative warehouse via logistics job
LogisticsPartner	TransportJob	Partner owns/operates vehicles for farm-to-warehouse transport only
GovernmentSubsidy	Farmer	Input subsidies (seed/fertilizer) disbursed to eligible farmers
CooperativeSociety	CooperativeFinancials	All income, costs, and profit/loss recorded in financials